Homeownership as Rental-Rate Risk Insurance*

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Abstract

We identify the stochastic process of individual rental rates by combining the PSID with external rent indexes, because a household panel alone cannot separate the persistence of rent risk from permanent cross-household heterogeneity: the two are near-observationally-equivalent at short panel lengths. Aggregate indexes point-identify the market component, which is at or near a unit root; netting it out identifies the idiosyncratic component, whose permanent variance is interior rather than at a boundary; and new-tenant versus sitting-tenant indexes measure the re-entry margin as a dispersed re-draw rather than a predictable discount. Carried through an estimated life-cycle tenure model, the process overturns the presumption that rent risk pushes households into owning: removing all rental-rate risk *raises* aggregate homeownership by 3.29 percentage points, because the dominant force is not the insurance value of owning but the lock-in value of cheap rental draws — claims that renters forfeit by buying. Renters holding expensive draws buy at twice the rate of those holding cheap ones, and the welfare incidence of rent risk is sharply bottom-loaded: removing it is worth $+6.8$ percent of lifetime consumption to the poorest quintile. A security-of-tenure policy that makes rental deals portable supplies the insurance directly, releases locked-in renters — raising ownership by 0.82 points rather than crowding it out — and delivers $+2.03$ percent to the poorest quintile. What looks like a preference for renting is, in substantial part, a portfolio position in rental claims.

JEL codes: D15, E21, R21.

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1 Introduction

The largest single price in most household budgets is the price of housing services. A renter pays that price on a spot market, where it can reset at every lease; an owner with a fixed-rate mortgage locks it in at purchase. Tenure choice is, among other things, a decision about how much of this price risk to bear — a risk-management decision as much as an investment one (Sinai and Souleles, 2005; Cocco, 2005; Rampini and Viswanathan, 2016). That homeownership might insure households against rental-rate risk is an old idea; the difficulty in evaluating it is that the object it turns on is rarely measured at the level at which the risk is actually borne: the stochastic process of the rent an individual household would pay for a given housing unit. The closest prior evidence infers the insurance motive indirectly, from how ownership rates co-move with rent volatility across cities (Amior and Halket, 2014). We instead measure the household-level rent process itself and trace it through an estimated model, which lets us separate the insurance value of owning from a countervailing force the cross-city approach cannot see — the value of holding on to a cheap rental claim.

This paper measures that process and traces it through an estimated life-cycle model of tenure choice. We make three points. The measurement requires combining a household panel with external rent indexes, because the panel alone cannot separate the persistence of rent risk from permanent cross-household heterogeneity. The measured process, carried through the model, implies that rental-rate risk barely moves *how many* households own — but reshapes *who* owns, who moves, and how lifetime welfare is distributed. And the model makes precise a policy substitution that the data motivate: security-of-tenure regulation and homeownership are alternative technologies for insuring the same risk.

Measurement. Within households that stay in the same unit, the residual rent decomposes into a permanent unit-and-household component, a persistent shock, and measurement error — the error-components machinery the income-dynamics literature developed to separate inequality from risk (Deaton and Paxson, 1994; Meghir and Pistaferri, 2004; Blundell et al., 2008; Storesletten et al., 2004), applied to the price of housing services. In a short household panel this decomposition is not identified: a flat autocovariance tail in the level of rents is consistent with a permanent household effect, with a near-unit-root shock, or with any mixture, and the two are observationally near-equivalent at the PSID's biennial frequency (Section 2). Unrestricted estimation returns the boundary corner — zero permanent variance, persistence pinned near one — rather than a point.

We break this identification failure with external rent indexes. An aggregate index carries no household fixed effect and is long, so the persistence of the *market* component of rents is point-identified by a time-series test, free of the short-panel problem: we estimate it on the regional CPI rent of primary residence back to 1977 and cannot reject a unit root in the national series and three of four census regions. Netting the market component out of household rents with region-by-

year fixed effects then identifies the *idiosyncratic* remainder within the panel, and its permanent variance moves off the boundary to an interior estimate. A third external contrast — the gap between new-tenant and sitting-tenant repeat-rent indexes — measures the re-entry margin: the rent a household re-draws when it moves. That gap grows slowly on average but is highly dispersed, so the re-entry margin is a risk about the level of the next rent, not a predictable discount. The result is a decomposition of rental-rate risk into a near-permanent market component, a persistent idiosyncratic component with genuine permanent heterogeneity, and a dispersed re-draw at moves; the estimated household rents track the official index within region almost one for one (slope 1.215). A by-product of the decomposition is a caution for calibration: measurement error accounts for roughly 68.0 percent of the variance of *reported* within-unit rent changes, so most of what moves in observed rents is noise, and the cross-sectional dispersion of rents is the slow accumulation of a persistent shock rather than year-to-year volatility. A model that reads rent risk off observed dispersion — the common practice — conflates permanent heterogeneity, persistent risk, and noise, and mismeasures both the size and the persistence of the risk households actually face.

Model. We embed this process in a life-cycle model in which households choose tenure, housing size, and saving, subject to a down payment, a mortgage-rate wedge, a moving cost, and the estimated income and house-price risks. The rent a renter faces evolves within a tenancy but is re-drawn when she moves to a new unit — the feature that turns ownership into insurance against re-entry risk and, at the same time, makes a cheap sitting rent worth holding on to. Insurance and lock-in are thus two sides of one mechanism rather than two separate assumptions. We estimate the model's four structural preference parameters by joint simulated minimum distance on five data moments; the discount factor is standard, the consumption weight is set to the observed renter rent-expenditure share, interest rates are measured from the data over the sample window, and all shock processes — including a family labor-income process whose permanent component enters as household types rather than risk — are read from the empirical estimates.

Three results follow. First, removing all rental-rate risk *raises* aggregate homeownership by 3.29 percentage points — the opposite of what a pure insurance story predicts. The sign identifies the dominant mechanism: rental-rate risk creates *claims*, and renters holding cheap draws stay renters to keep them; a Shapley decomposition attributes 2.74 of the 3.29 points to the idiosyncratic component whose dispersion generates those claims, while the pure-uncertainty (insurance) margin is small (-0.36 points) and of the sign insurance logic predicts. Second, the welfare consequences are large and sharply distributional: the consumption-equivalent value of removing rent risk ranges from -9.4 percent at the tenth percentile to $+4.7$ percent at the ninetieth, is largest — $+6.8$ percent — for the poorest households, and is largest across rent positions — $+8.05$ percent — for those stranded on expensive draws. Third, the lock-in margin is visible in behavior: a renter's probability of buying rises from 0.29 to 0.51 across the current rent draw, and making rental deals portable cuts the mobility of households holding expensive draws from 0.39 to 0.15 — a large share of

renter mobility is moving to re-draw the rent.

Policy and contribution. These mechanisms have a direct policy reading, which we pursue in partial equilibrium. Full portability of rental deals — a stylized security-of-tenure guarantee — *raises* ownership by 0.82 points while delivering $+0.71$ percent of consumption on average and $+2.03$ percent to the poorest quintile: it supplies the insurance directly and releases the locked-in renters whom cheap draws hold back from buying. Tenant protection, in this economy, is not a rival to homeownership but a complement to it — a sharper claim than the substitution the rent-control literature has debated on other margins (Diamond et al., 2019; Sommer and Sullivan, 2018). Our contribution is threefold. Empirically, we identify household rental-rate risk by combining a panel with external indexes, a strategy that resolves an identification problem internal to the panel and that we validate against the official series. Structurally, the move-contingent rent re-draw unifies the insurance and lock-in motives that the housing literature has treated separately, and we quantify both inside one estimated model. Substantively, we show that rental-rate risk is better understood as an asset households hold — a claim on a cheap unit — than as a shock that pushes them toward owning: its removal raises ownership, and its burden falls on the poor and the unlucky, not on tenure per se.

Related literature. The paper connects three strands. It brings the income-dynamics decomposition (Deaton and Paxson, 1994; Storesletten et al., 2004; Blundell et al., 2008) to housing costs, where the analogue of the permanent-versus-transitory question is the persistence of within-unit rent risk, and where — unlike income — a short panel cannot answer it without external data. It adds to the life-cycle housing literature (Cocco, 2005; Li et al., 2016; Bajari et al., 2013; Halket and Vasudev, 2014; Banks et al., 2015; Ejarque, 2014; Sommer and Sullivan, 2018; Kaplan et al., 2020) a rent process identified from the data rather than calibrated to cross-sectional dispersion, and a lock-in margin that a mean-preserving change in rent dispersion activates — a margin distinct from the tenure lock-in that transaction costs and house-price risk generate in that literature. And it speaks to the housing-policy literature (Diamond et al., 2019; Favilukis et al., 2017) by measuring the insurance a security-of-tenure policy would provide and the ownership it would, or would not, displace.

The paper proceeds as follows. Section 2 identifies and estimates the rental-rate process. Section 3 sets out and estimates the model. Section 4 reports the decomposition, welfare, and policy experiments and their robustness. Section 5 concludes.

2 Identifying Rental-Rate Risk

2.1 The Object: Four Margins of Rental-Rate Risk

The rental-rate risk a household bears decomposes along two dimensions: whether the movement is common to the local market or specific to the rented unit, and whether it arrives while the household sits in its unit or when it re-enters the rental market. The four cells require different measurements and carry different insurance values, and the lifecycle model of Section 3 takes each as an input: the persistence and innovation size of the market component, the dynamics of the idiosyncratic unit component within a tenancy, the size of the re-draw at re-entry, and the permanent household-level dispersion in rent levels.

2.2 Data

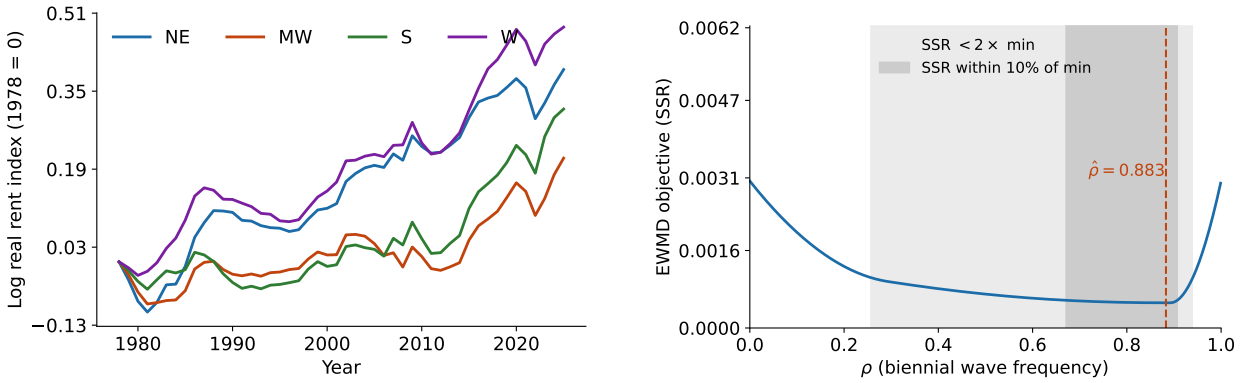
Household rents come from the PSID family files, 2001–2021 (21,319 renter observations from 6,108 households), deflated by the CPI all-items-less-shelter index so that rent movements are measured against non-housing purchasing power. Market rent indexes come from three external sources, frozen with checksums at download: the CPI rent of primary residence by census region (monthly, from December 1977 — 48 annual observations), the BLS new-tenant and all-tenant repeat-rent research indexes (quarterly; all-tenant from 1999, new-tenant from 2005), and the Zillow observed rent index (metro level, 2015 onward, reserved for cross-metro dispersion once the restricted geocode permits a metro match).

2.3 Why a Household Panel Is Not Enough

Within stayer spells the rent residual obeys $u_{it} = f_i + z_{it} + m_{it}$, where f_i is a household fixed effect, z_{it} an AR(1) with stationary variance σ_z^2, and m_{it} i.i.d. measurement error. In levels, $\mathrm{Cov}(u_t, u_{t+k}) = \sigma_f^2 + \sigma_z^2 \rho^k$: a flat autocovariance tail is consistent with a household fixed effect, with persistence near one, or with any mixture. At the PSID's biennial frequency and spell lengths the tail hardly decays (lag-3 to lag-1 ratio 0.894), and unrestricted estimation returns the boundary corner $\hat{\sigma}_f^2 = 0$ with $\hat{\rho}$ near the random-walk end — a ridge, not a point. No within-panel specification resolves this: the two parameters are nearly observationally equivalent at short T.

2.4 External Identification of the Market Component

An aggregate rent index removes both obstacles at once: it carries no household fixed effect, and it is long. We estimate an annual AR(1) on the real log rent index of each census region (January–June averages, matching PSID interview timing), correct the well-known downward bias of least squares near the unit root by inverting the simulated median function of the estimator at the exact sample size (Andrews, 1993), and test for a unit root under a decision rule fixed before estimation. The rule binds us: a rejection reports the median-unbiased persistence with its interval; a failure to reject commits the market component to a unit root with drift, reported as an interval, with no



(a) External market rent, by region

(b) The identification ridge

Figure 1: External identification. Left: real market rent by census region (CPI rent of primary residence deflated by CPI less shelter), the long aggregate series whose persistence is point-identified. Right: the objective of the within-panel error-components estimator against the persistence parameter ρ — flat over a wide range, the near-non-identification that the external series resolves; the shaded band is the set within ten percent of the minimum.

Table 1: Persistence of real market rent growth: regional and national CPI rent indexes.

	Sample	$\hat{\rho}_{\text{OLS}}$	$\hat{\rho}_{\mu}$	90% CI	ADF p	$\sigma^{(3)}$	Verdict
Northeast	1978–2025	0.729	0.847	[0.642, 1.020]	0.028	0.034	trend-stationary
Midwest	1978–2025	0.849	1.020	[0.809, 1.020]	0.443	0.038	unit root
South	1978–2025	0.937	1.020	[0.961, 1.020]	0.948	0.040	unit root
West	1978–2025	0.888	1.020	[0.870, 1.020]	0.439	0.040	unit root
National (full)	1941–2025	0.931	1.020	[0.920, 1.020]	0.973	0.046	unit root
National (1978 window)†	1978–2025	0.842	1.020	[0.799, 1.020]	0.644	0.038	unit root

Notes. Real market rent growth by census region and the nation; series construction and estimation method in Appendix B. $\hat{\rho}_{\text{OLS}}$: least-squares AR(1) coefficient (constant and trend); $\hat{\rho}_{\mu}$: Andrews (1993) median-unbiased estimate with 90% interval, with 1.020 the inversion-grid ceiling. ADF p: augmented Dickey–Fuller p-value. $\sigma^{(3)}$: three-year innovation standard deviation. †Governing series, fixed before estimation.

re-specification. Table 1 reports the results. The unit root is not rejected in three of four regions and in the national series ($p = 0.644$); only the Northeast rejects ($p = 0.028$), with median-unbiased persistence 0.847. The market rent component therefore enters the model as a unit root with a three-year innovation standard deviation of 0.038; its deterministic drift is absorbed by the level normalization described in Section 3.

2.5 The Idiosyncratic Component

With the market identified externally, the within-household step changes in three ways, fixed in advance, relative to the earlier draft: region-by-year fixed effects replace national year effects in the hedonic, so all region-common movement is absorbed nonparametrically; rents are deflated by the region-matched less-shelter index; and the error-components parameters are estimated by equal-weighted minimum distance on seven autocovariance moments, replacing a two-step weighting

Table 2: Rent error-components estimates: naive within-panel versus external identification.

	Naive year FE, two-step	Naive scale-free	External ID region×year FE, EWMD
$\rho^{(2)}$ (wave)	0.905	0.959	0.883
ρ (3-yr model)	0.861	0.939	0.830
σ_η (3-yr model)	0.279	0.179	0.285
σ_z^2	0.302	0.269	0.261
σ_f^2	0.000	0.000	0.018
σ_m^2	0.064	0.066	0.061
ME share of level variance (%)	17.0	–	17.3
ME share of FD variance (%)	69.2	71.0	68.0
Tail ratio $\mathrm{lv}_3/\mathrm{lv}_1$	0.894	–	0.867
Bootstrap J p-value	–	–	0.122

Notes. Error-components estimates $u_{it} = f_i + z_{it} + m_{it}$, $z_{it} = \rho^{(2)} z_{i,t-1} + \eta_{it}$, on strict renter stayer spells. The first two columns are the naive within-panel protocol (national year effects, all-items deflator, two-step weighting and its scale-free first step); the last is the external-identification estimation (region×year effects, less-shelter deflator, equal-weighted minimum distance). Method in Section 2 and Appendix C. Tail ratio: lag-3 to lag-1 level autocovariance. J: bootstrap overidentification p-value.

that finite-sample evidence (Altonji and Segal, 1996) counsels against. Because any deflator that varies only at the region-year level is absorbed by the fixed effects, the moments are numerically invariant to the deflator swap — a built-in check of the design that holds to machine precision.

Table 2 reports the decomposition; the first two columns show, for comparison, what the naive within-panel protocol returns. The unit fixed-effect variance moves off the boundary to an interior $\sigma_f^2 = 0.018$; persistence is 0.883 at the biennial frequency and is set-identified: the profile-distance interval within ten percent of the minimum spans $[0.671, 0.907]$. The innovation standard deviation is pinned at 0.285 per three-year period. The estimated region-year effects track the official index: regressing them on the external series gives a slope of 1.215 (s.e. 0.095) with $R^2 = 0.81$.

Two features of the estimates matter for how rent risk should enter a model. First, most of the movement in *reported* rents is noise: measurement error accounts for 68.0 percent of the variance of within-unit rent changes, and an overidentification test of the classical measurement-error specification does not reject ($p = 0.122$). Second, what survives is persistent: the innovation is nearly permanent, so the cross-sectional dispersion of rents is the slow accumulation of a durable shock rather than year-to-year churning. Together these imply that a calibration reading rent risk off observed cross-sectional dispersion — a common shortcut — conflates three distinct objects, permanent heterogeneity, persistent risk, and measurement error, and mismeasures both the size of rental-rate risk (upward, by counting noise) and its persistence (with no basis to place it). The model of Section 3 takes the decomposed quantities as inputs precisely to avoid this.

Table 3: The rent re-draw at moves: sitting-tenant drift (W1) and household-level dispersion (W2).

Panel A: external ATR–NTR wedge (W1, selection-free)	Quarterly	Annualized
Drift of gap δ	0.000335	0.0013
Volatility of Δgap	0.010	0.020
Drift, 2010q1–2019q4	0.001098	0.0044
Volatility, 2010q1–2019q4	0.008	0.015
Gap level, 2025q3	0.027	
Cumulative divergence $\delta \times L$: L = median tenancy	0.0027	
L = model period (3 yr)	0.0040	
L = mean tenancy	0.0041	
Panel B: PSID re-entry dispersion (W2), Δ v2 residual	Movers	Stayers
Mean	0.021	-0.037
Standard deviation	0.641	0.424
ME-corrected re-draw s.d.	0.538	–
10th percentile	-0.598	-0.340
Median	0.009	-0.041
90th percentile	0.655	0.324
Observations	6,099	3,823
Mover $-$ stayer mean	0.058	
Lee bounds on mover mean	$[-0.138, 0.178]$	
Exit-to-ownership share q	0.138	
Renter tenancy: median / mean (years)	2	3.1

Notes. Panel A: log gap between the BLS new-tenant and all-tenant rent indexes, quarterly, normalized to zero at 2005q1; deflator-free by construction. Drift and volatility are annualized; $\delta \times L$ is the sitting-tenant discount forfeited over an L-year tenancy. Panel B: change in the rent residual between consecutive renter-to-renter PSID waves, movers versus strict within-spell stayers, weighted; Lee bounds trim on renter exits to ownership. Method, and the measurement-error correction of the mover standard deviation to the model-relevant re-draw dispersion, in Section 2 and Appendix F.

2.6 The Re-Entry Margin

The BLS new-tenant and all-tenant indexes measure the two sides of the re-entry margin without household selection: their gap is free of any common deflator by construction, and it isolates exactly the wedge that sitting-tenant rent rigidity opens between what a continuing tenant pays and what a new tenant would (Genesove, 2003). New-tenant rents outgrow sitting-tenant rents by only 0.13 percent per year, so the sitting discount forfeited over a median tenancy is economically negligible. The dispersion is not: across PSID moves, the standard deviation of residual rent changes is 0.641 against 0.424 within spells, and the mover mean is bounded by selection — trimming on exits to ownership gives $[-0.138, 0.178]$ — so its sign is not identified. The re-entry margin is a dispersion phenomenon, not a level phenomenon: what ownership insures is the draw, not a discount. Table 3 collects these results, and the stayer-selection sensitivity of the within-spell moments is bounded in the same trimming spirit ($[0.006, 0.310]$ for the first-difference variance).

2.7 From Estimates to Model Inputs

The model receives the market component as a unit root with the external innovation size; the idiosyncratic component as the estimated AR with its interior fixed-effect variance entering as permanent rent-level types; measurement error excluded from the shock processes; and the re-entry margin as a partial re-draw whose correlation with the previous draw is implied by the certified mover variance. The identification asymmetry that motivated this design is worth recording: house values, estimated with the same machinery, separate unit heterogeneity from persistence within the panel; rents do not, which is precisely why the external instruments carry the market cell of the decomposition.

3 A Life-Cycle Model of Tenure Choice

We embed the estimated rental-rate process in a partial-equilibrium life-cycle model of tenure choice. A household chooses each period whether to rent or own, how large a dwelling to occupy, how much to consume, and how much to save, facing idiosyncratic risk in labor income, in the rent it would pay as a renter, and in the price of owner-occupied housing. Renting exposes the household to the full rental-rate process of Section 2; owning fixes the price of housing services at the cost of a down payment, a mortgage-rate wedge, and a transaction cost on adjustment. The model's purpose is to measure how each component of rental-rate risk maps into tenure, mobility, and welfare, and to evaluate policies that alter the risk households bear.

3.1 Demographics and Preferences

Households enter at age 27 and live for $T = 18$ three-year periods, to age 81. A household is described at the start of period t by its tenure $o_t \in \{0, 1\}$ (with $o_t = 1$ for owners), the size $h_t \in \mathcal{H}$ of the dwelling it occupies, its net financial assets a_t, and the exogenous states governing income, rent, and house prices introduced below. Each period the household chooses next-period tenure $o_{t+1} \in \{0, 1\}$, dwelling size $h_{t+1} \in \mathcal{H}$, non-durable consumption $c_t \geq 0$, and next-period assets a_{t+1}.

Flow utility is constant-relative-risk-aversion over a constant-elasticity aggregate of consumption and housing services, with a tenure-specific quality shifter:

$$u(c, h, o) \;=\; \frac{\left[\alpha\, c^{\nu} + \psi_o\,(1-\alpha)\, h^{\nu}\right]^{\frac{1-\gamma}{\nu}}}{1-\gamma}, \qquad \psi_1 = 1, \;\; \psi_0 \in (0, 1), \tag{1}$$

where γ is the coefficient of relative risk aversion, $1/(1-\nu)$ the elasticity of substitution between consumption and housing, α the consumption weight, and $\psi_0 < 1$ a quality wedge that makes a unit of rented housing services worth less than a unit of owned services — capturing the control, customization, and security differences between the two tenures in reduced form. At the end of

life the household values terminal net worth through a warm-glow bequest,

$$B(a,h,o,p) \;=\; \theta_B \,\frac{\left(a + o\,p\,h\right)^{1-\gamma}}{1-\gamma}, \tag{2}$$

with weight $\theta_B \geq 0$, where p is the terminal house price so that an owner bequeaths financial assets plus home equity. Future utility is discounted at rate β per period.

3.2 Income, Rents, and House Prices

The household faces three exogenous risks. Log labor income is the sum of a deterministic age profile, a persistent component, and a transitory component,

$$\log y_{it} \;=\; \chi_t \;+\; f_i^{y} \;+\; \omega_{it} \;+\; \varepsilon_{it}, \qquad \omega_{it} = \rho_y\,\omega_{i,t-1} + \eta_{it}^{y}, \tag{3}$$

where χ_t is the estimated age-earnings profile (Appendix D), f_i^{y} a permanent income type — three mean-one values carrying the estimated permanent variance 0.082, known to the household and hence identity rather than risk — $\rho_y = 0.829$ with persistent innovation standard deviation 0.326 per period, and ε_{it} the transitory component (0.250). The persistent state is initialized at age 27 at its observed 27–31 dispersion rather than the stationary one. The log rent a household would pay for its unit is the three-component process identified in Section 2,

$$\log q_{it} \;=\; g_t \;+\; z_{it} \;+\; f_i, \tag{4}$$

in which g_t is a common market component, z_{it} an idiosyncratic unit-and-household component, and f_i a permanent rent-level type. The market component is a random walk,

$$g_t \;=\; g_{t-1} \;+\; \eta_t^{g}, \qquad \eta_t^{g} \sim (0, \sigma_g^2), \tag{5}$$

with $\sigma_g = 0.0379$ per period, discretized on an age-expanding grid whose level multipliers are renormalized to mean one at every age, so that shutting the component down in a counterfactual leaves the cross-sectional mean of rents unchanged by construction. The idiosyncratic component is a stationary autoregression, $z_{it} = \rho_q\, z_{i,t-1} + \eta_{it}^{z}$, with $\rho_q = 0.830$ and $\sigma_{\eta,q} = 0.285$ per period; the permanent type f_i takes three values (factors 0.78 to 1.25), drawn at birth and fixed thereafter, so that f_i is household heterogeneity rather than risk. Together with the three income types f_i^{y} of (3), drawn independently at birth, households come in nine permanent configurations; both type dimensions are identity, not uncertainty, and carry no precautionary motive. The transition of z is *move-contingent*: a household that keeps its rental unit carries z forward by its estimated autoregression, but a household that enters a *new* rental unit — because it was an owner, or because it

changes size — re-draws z from a transition with correlation $\rho_{\text{move}} = 0.446$ to its previous value,

$$z_{i,t+1} \sim \begin{cases} \Pi_{\text{stay}}(\cdot \mid z_{it}) & \text{if the household keeps its unit,} \\ \Pi_{\text{move}}(\cdot \mid z_{it}) & \text{if it enters a new rental unit,} \end{cases} \tag{6}$$

where both transitions share the grid and the stationary distribution of z, so the re-draw preserves the cross-sectional dispersion of rents while partially resetting a household's own draw. This single feature is what makes owning insurance against re-entry risk and, at the same time, makes a cheap sitting rent worth holding: the derivation of ρ_{move} from the data is in Appendix F. House prices follow a stationary autoregression, $\log p_t = \rho_p \log p_{t-1} + \eta_t^p$, with $\rho_p = 0.650$ and $\sigma_{\eta,p} = 0.327$ per period (Appendix E). All level grids are normalized so that mean income, the mean rent-to-income ratio, and the mean price-to-income ratio match their PSID values at age 27.

3.3 Assets, Housing, and Frictions

Housing is chosen from a discrete menu $\mathcal{H}$ of sizes. A renter pays the current rent q_t per unit; a buyer pays the current price p_t per unit and may finance the purchase with collateralized debt. Net financial assets earn an interest schedule with a *mortgage wedge*,

$$r(a) = \begin{cases} r_s & \text{if } a \geq 0, \\ r_m & \text{if } a < 0, \end{cases} \qquad r_m > r_s, \tag{7}$$

with both rates measured from the data: over the estimation window the real 30-year mortgage rate averaged 2.72 percent per year and the real return on liquid saving 0.96 percent (Section 3.7); only owners may borrow, and only up to a fraction $1 - \phi$ of the value of the house, so a buyer posts a down payment of at least the share ϕ. Owner-occupied housing depreciates at rate δ per period. Changing dwelling — whether by size or by tenure — carries a proportional transaction cost ζ on the value of the departing unit: broker fees, moving costs, and the friction of search. The cost applies to any physical move, so a tenure switch at unchanged size is priced like any other move; this is the same event that triggers the rental re-draw of (6), keeping the cost and expectation sides of the problem consistent. The friction generates an inaction region in the size choice.

3.4 The Household's Problem

Collect the exogenous states in $s_t = (g_t, z_t, f, p_t, y_t)$ and write the endogenous states as (a_t, h_t, o_t). Consider a household that has chosen next-period tenure $o' = o_{t+1}$. Given that choice it selects dwelling size and saving to solve

$$W_t(s_t, a_t, h_t, o_t;\, o') = \max_{h' \in \mathcal{H},\, a'} \left\{ u(c, h', o') + \beta\, \mathbb{E}_t\, V_{t+1}(s_{t+1}, a', h', o') \right\}, \tag{8}$$

subject to the budget identity, the borrowing constraints, and non-negative consumption:

$$c + P_t^{o'} h' + \zeta P_t^{o_t} h_t \mathbf{1}\{h' \neq h_t \vee o' \neq o_t\} + a' = y_t + a_t\big(1 + r(a_t)\big) + o_t\, p_t\, h_t(1-\delta), \tag{9}$$

$$a' \geq -(1-\phi)\, p_t\, h' \mathbf{1}\{o' = 1\}, \qquad c \geq 0, \tag{10}$$

where $P_t^1 = p_t$ and $P_t^0 = q_t$ are the purchase and rental prices of a unit of housing, and the second line imposes the collateral limit for buyers and $a' \geq 0$ for renters. Equation (9) says that cash-on-hand — labor income, financial assets with interest, and, for a current owner, the depreciated resale value of the house — is divided among consumption, the cost of the next-period dwelling, the transaction cost on any size change, and saving. The expectation in (8) is taken over next period's exogenous states, with the rent component z evolving by the move-contingent law (6) according to whether the choice (o', h') keeps the household in its unit or places it in a new one.

The tenure choice itself is subject to an additive taste shock. Let the two choice-specific values be $W_t(\cdot;0)$ and $W_t(\cdot;1)$; the household draws a pair of Type-I extreme-value taste shocks with scale ξ and chooses the tenure with the higher shock-inclusive value. Integrating over the shocks gives the expected value and the tenure probability in closed form,

$$V_t(s_t, a_t, h_t, o_t) = \xi \left[\gamma_E + \log\big(e^{W_t(\cdot;0)/\xi} + e^{W_t(\cdot;1)/\xi}\big)\right], \tag{11}$$

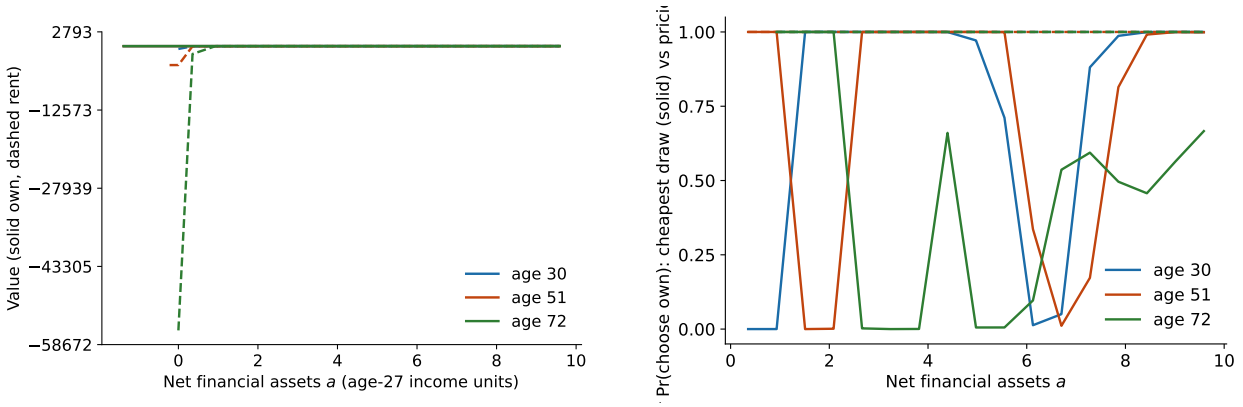
$$\Pr(o_{t+1} = 1 \mid s_t, a_t, h_t, o_t) = \frac{e^{W_t(\cdot;1)/\xi}}{e^{W_t(\cdot;0)/\xi} + e^{W_t(\cdot;1)/\xi}}, \tag{12}$$

with γ_E the Euler–Mascheroni constant. The smoothing applies to the binary tenure margin only; the continuous choices h' and a' are solved by exact maximization in (8). At the terminal date the continuation value is the bequest (2) evaluated at the chosen terminal position, $V_{T+1} = B(a_{T+1}, h_{T+1}, o_{T+1}, p_{T+1})$. Equations (8)–(12) define the dynamic program the household solves.

3.5 Equilibrium

The environment is partial equilibrium: the rent and house-price processes are exogenous, taken as estimated in Section 2, and housing supply is perfectly elastic at those prices.

Definition 1 (Recursive equilibrium)**.** Given the price processes for q and p and the income process for y, a recursive equilibrium is a collection of value functions $\{V_t, W_t\}_{t=1}^{T}$ and policy functions $\{h'_t, a'_t, \Pr(o_{t+1} = 1)\}_{t=1}^{T}$ that solve (8)–(12) at every state, together with the cross-sectional distribution over states that these policies induce, period by period, from the age-27 initial conditions. These are drawn from the PSID's empirical joint distribution of wealth and tenure at 27 — each simulated household inherits an observed household's tenure and its financial assets, with home equity netted out of owners' wealth so the house is not counted twice; initial wealth heterogeneity, in particular the near-zero wealth of most young renters, is thus a measured input rather than a calibration choice.



(a) Value functions

(b) Lock-in: renter's probability of buying

Figure 2: Solved value and tenure policies at the estimated parameters. Left: value of owning (solid) and renting (dashed) against net financial assets, ages 30/51/72, median shocks, middle rent type. Right: a renter's probability of buying against assets, for the cheapest (solid) versus the priciest (dashed) current rent draw — the cheap-draw renter delays buying where the pricey-draw renter buys.

No market-clearing condition is imposed: because we hold the joint rent and price processes fixed at their measured values, the model isolates how households reallocate tenure, mobility, and consumption in response to the risk those processes carry, without feeding tenure demand back into prices. A general-equilibrium closure — in which rents and prices respond to aggregate tenure demand, adding one market-clearing condition per housing market — would change the level effects of the policy experiments and is left for future work; the risk decomposition, which is the paper's object, is a partial-equilibrium statement.

3.6 Model Mechanisms

Before turning to estimation we illustrate the two forces the model isolates, using the solved value and policy functions at the estimated parameters. Figure 2a plots the value of owning and of renting against financial assets, by age. Both rise and are concave in wealth; the owner–renter gap is positive and narrows as assets accumulate, since a wealthy renter can self-insure against rent risk through savings while a poor renter cannot — ownership is worth most to the asset-poor. Figure 2b makes the lock-in force visible: it plots a renter's probability of buying against assets, separately for a household holding the cheapest current rent draw and one holding the most expensive. At the same wealth, the household with an expensive draw buys with near-certainty while the one with a cheap draw delays — the cheap draw is a valuable claim the household is reluctant to give up by moving or buying. The same figure is the counterpart of the moving cost: Figure 3 shows the (S, s) inaction region in the size choice, the band of states over which an owner keeps her current dwelling rather than pay the transaction cost to resize; the band widens with the moving cost, as a comparative static confirms.

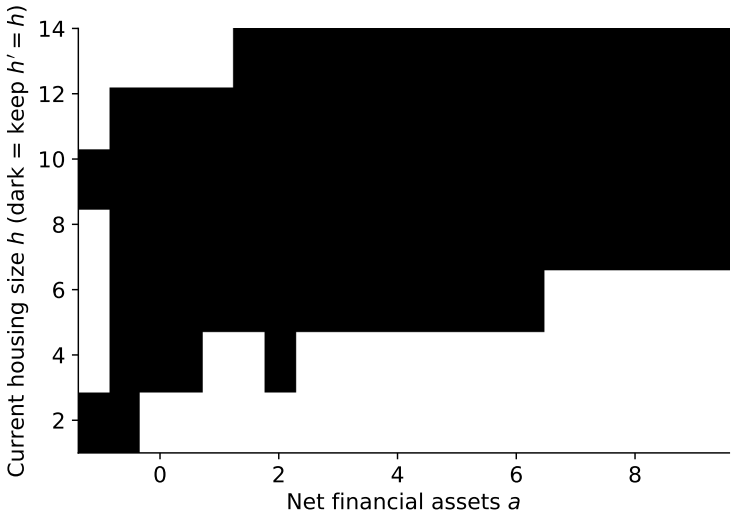


Figure 3: The (S, s) inaction region in housing size. Shaded: states (a, h) at which an owner keeps her current dwelling ($h' = h$) rather than pay the transaction cost to resize; the band widens with the moving cost ζ.

3.7 Calibration and Estimation

The model's parameters divide into an externally set block and an estimated block (Table 4). Externally set are the objects for which either a convention or a direct data counterpart exists. The discount factor is the standard annual 0.96. The interest rates come from the data over the 2001–2021 estimation window: the mortgage rate is the average Freddie Mac 30-year fixed rate less realized CPI inflation, 2.72 percent per year in real terms, and the savings rate is the average 10-year Treasury yield less the same inflation, 0.96 percent; the implied borrowing–saving wedge is 1.76 points per year, in the range conventional in the housing literature. The consumption weight is set externally to $\alpha = 0.80$, one minus the median renter rent-expenditure share of 0.206: the renter's intratemporal condition, $q_t h/c = [(1-\alpha)\psi_0/\alpha]\,(h/c)^{\nu}$, reduces to exactly this mapping only in the Cobb–Douglas limit, but at the estimated ν the approximation is close and the share-based value is transparent. The tenure taste scale ξ is fixed at 0.005: it is a smoothing device for the discrete choice (Iskhakov et al., 2017), not a structural taste, and it is set small enough that tenure is effectively a deliberate decision — halving or doubling it left every headline object unchanged in the sensitivity sweeps run during model development. The down payment, depreciation, and moving cost take standard values, and the three income, rent, and house-price processes are the empirical estimates of Section 2 and Appendices D–E, not re-tuned.

Four structural parameters are estimated: the renter housing-weight wedge ψ_0, the bequest weight θ_B, risk aversion γ, and the CES exponent ν. Estimation is joint simulated minimum distance: the four parameters minimize the equal-weighted sum of squared relative distances between five simulated and data moments — aggregate homeownership, homeownership at age 30, the owner median wealth-to-income ratio, the renter median rent-to-income ratio, and the renter move rate

— with all parameters moving all moments; the system is overidentified by one moment and we make no moment-by-moment identification claims. The model is solved exactly and a panel of 100,000 lives is simulated with common random numbers; no likelihood is evaluated and no conditional-choice-probability inversion is used. Because the objective — built from simulated medians and transition frequencies of a discrete-choice model — is not differentiable, we do not rely on a local optimizer: parameters are selected by a global quasi-random (Sobol) scan of the economically admissible box ($\psi_0 \in [0.4, 0.9]$, $\theta_B \in [0.2, 3]$, $\gamma \in [1.1, 3]$, $\nu \in [0.05, 0.6]$), every point evaluated at the same grid resolution at which all results are reported, and the minimizing point is adopted. The scan's value is transparency: the full evaluated surface is retained, so the reported parameters are the best of an inspectable table rather than the endpoint of an untraceable descent path.

Parameters. Table 4 collects every parameter of the model in four blocks: estimated preferences, externally set preferences, prices and frictions, and the shock processes estimated outside the model. The four estimated values each carry economic content worth stating. The renter housing weight is $\psi_0 = 0.431$: a unit of rented housing delivers a bit less than half the service flow of the same unit owned — a moderate wedge bundling quality differences between the rented and owned stocks with the non-pecuniary premium on owning, far from the order-of-magnitude penalties that appear when tenure models are forced to rationalize ownership with preferences alone. Risk aversion is $\gamma = 2.33$, inside the conventional one-to-three range of the consumption-saving literature. The bequest weight $\theta_B = 2.150$ implies old households value terminal wealth strongly enough to hold housing to the end of life, consistent with the near-absence of old-age tenure exits in the data. The CES exponent $\nu = 0.60$ implies an elasticity of substitution between housing and consumption of $1/(1-\nu) = 2.5$ — above the range most calibrations use (0.5 to 1.25); we report this plainly rather than restrict it: it is the value the renter expenditure and mobility moments jointly ask for, and flattening it is a natural robustness exercise for the re-run sweep. Among the external settings, only δ merits comment: 0.12 per three-year period (≈ 4 percent annually) should be read as the all-in cost of holding a home — depreciation, maintenance, property taxes, insurance — not depreciation alone.

Fit. Tables 5 and 6 report the targeted and untargeted fit separately, each with its signed relative gap. On the targeted side, aggregate homeownership is matched within four percent and no moment misses by more than a quarter; the two largest gaps — age-30 ownership too high, renter mobility too low — are the model's genuine tensions, not artifacts of the moment set. The untargeted side is where the model earns or loses credibility, and the ledger is mixed in an informative way. It earns: the life-cycle profile in Figure 4 rises and is concave with no age moment beyond 30 in the objective — entry is genuinely delayed by young households who start at the data's initial wealth and save toward ownership — and the house-value-to-income ratio of owners comes within sixteen percent. It loses, in a correlated cluster: owners deleverage too fast (a mortgaged

share of 0.393 against 0.693, home equity correspondingly high, non-housing wealth near zero), old households run wealth down where the data show them holding it, and renters still carry too much wealth. These margins share one plausible root: mortgage repayment in the model is unrestricted, so owners prepay immediately rather than amortize — a documented extension, not a free parameter, and the first candidate for the next model revision.

Table 4: Parameters: estimated and externally set.

Parameter	Interpretation	Value	Source
Estimated preferences *(global SMM scan, Table 14)*			
ψ_0	renter housing weight (owner = 1)	0.431	estimated jointly
θ_B	bequest weight	2.150	estimated jointly
γ	relative risk aversion	2.33	estimated jointly
ν	CES exponent	0.60	estimated jointly
Externally set preferences			
α	consumption weight	0.80	1− median renter rent share
β	discount factor (annual)	0.96	standard
ξ	tenure taste scale	0.005	smoothing device
Prices and frictions			
r_s	saving rate, real %/yr	0.96	10-yr Treasury − CPI
r_m	mortgage rate, real %/yr	2.72	30-yr FRM − CPI
φ	down-payment share	0.20	standard
δ	depreciation + holding costs (3-yr)	0.12	standard
ζ	moving cost (share of unit value)	0.08	standard
Shock processes *(estimated outside the model)*			
rent	market RW + AR(1) + types + re-draw	—	Section 2
income	profile + types + AR(1) + transitory	—	Appendix D
house price	AR(1) + fixed effect	—	Appendix E

Notes. The four preference parameters minimize the equal-weighted sum of squared relative gaps between the five simulated and data moments of Table 5, jointly and overidentified by one moment; the selection is the global Sobol scan of Table 14. Interest rates are 2001–2021 averages: Freddie Mac 30-year fixed-rate mortgage (4.85 percent nominal) and 10-year Treasury (3.09 percent), each less realized CPI inflation (2.13 percent). The consumption weight is one minus the median renter rent-expenditure share (0.206). No likelihood or CCP inversion is used; the solver is exact dynamic programming.

Table 5: Model fit I: targeted moments.

Moment	Data	Model	Gap (%)
Homeownership, aggregate	0.731	0.758	+3.7
Homeownership, age 30	0.537	0.679	+26.4
Owner median wealth-to-income	2.707	3.186	+17.7
Renter median rent-to-income	0.206	0.242	+17.5
Renter move rate (3-yr)	0.563	0.435	−22.7

Notes. The five moments enter the joint SMM objective with equal weights on squared relative gaps; the gap column is the signed relative deviation of model from data. Beyond the aggregate level and age 30, the life-cycle ownership profile is untargeted (Figure 4).

Table 6: Model fit II: untargeted moments.

Moment	Data	Model	Gap (%)
Portfolios and credit			
Share of owners with a mortgage	0.693	0.393	−43.3
Owner median home-equity-to-income	1.287	2.531	+96.7
Owner median non-housing wealth-to-income	0.867	0.000	−100.0
Owner median house-value-to-income	2.478	2.882	+16.3
Renter median wealth-to-income	0.106	2.644	+2394.3
Tenure dynamics and dispersion			
Own→rent rate (3-yr)	0.064	0.104	+62.5
Rent→own rate (3-yr)	0.234	0.378	+61.5
Ownership age gradient (55+ − ≤35)	0.311	0.123	−60.5
Median wealth, ages 70–78 / 50–58	2.084	0.667	−68.0
SD log rent paid (net of meas. error)	0.541	0.643	+18.9

Notes. No moment in this table enters the estimation objective. The panels report the margins that discriminate between mechanisms: portfolio composition and the credit margin, tenure transitions, the life-cycle gradient, and the pass-through of rent dispersion into rents paid. Remaining gaps are discussed in the text.

The model is solved by backward induction with exact discrete maximization over the (a', h') grid; the grid resolution, the property tests every solution must pass, and the welfare computation are described in Appendix G.

4 Insurance, Lock-In, and Policy

Every counterfactual in this section is labeled by the question it answers. Type-A exercises hold preferences fixed and remove a risk component mean-preservingly, with common random numbers pairing each household across worlds: they measure what the estimated households would pay for, or lose from, the component's removal; Experiment A″ extends the same construction across all three risk sources at once. The Type-B exercise removes all rental-rate risk and re-anchors

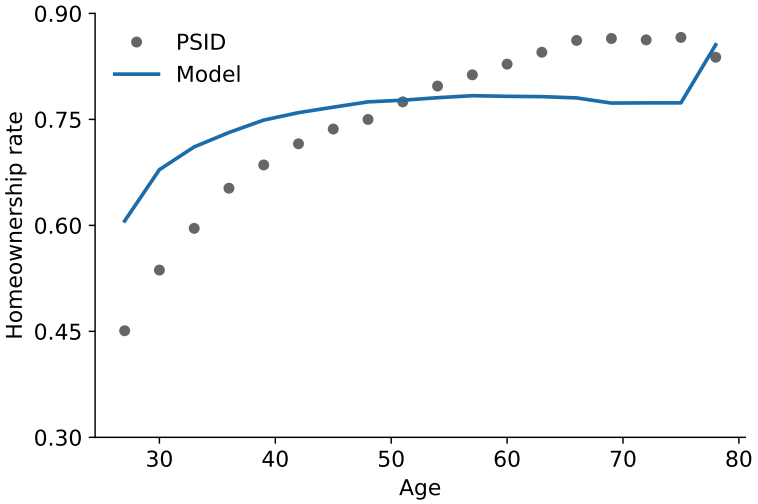


Figure 4: Homeownership over the life cycle: model at the estimated parameters (line) versus PSID (markers).

the renter quality wedge to baseline ownership: the resulting parameter shift measures how much of the estimated distaste for renting stands in for priced rent risk. Type-C exercises change policy instruments on the household's constraint set at fixed prices, and carry the partial-equilibrium caveat explicitly. Each experiment is reported in its own table (Tables 7–11).

Decomposition (A). Removing all rental-rate risk *raises* aggregate homeownership by 3.29 percentage points. The Shapley attribution locates the force: 2.74 points come from the idiosyncratic component, against 0.38 from the re-entry re-draw and 0.16 from the market unit root. And the sign identifies the mechanism: if ownership were primarily insurance against rent risk, removing the risk would remove the reason to own and the rate would *fall*. Instead it rises — because what rental-rate risk mainly creates in this economy is *claims*: renters holding cheap idiosyncratic draws own an asset they forfeit by buying, and stay renters to keep it. The pure-insurance margin, isolated by removing uncertainty while keeping the cross-sectional dispersion of draws, is -0.36 points — small, and of the sign insurance logic predicts. Rental-rate risk suppresses homeownership through lock-in, not despite insurance.

The welfare consequences are large and sharply redistributive. The mean consumption-equivalent value of removing all rent risk is $+1.08$ percent, but the ex-post incidence runs from -9.4 percent at the tenth percentile to $+4.7$ at the ninetieth; the poorest lifetime-consumption quintile gains $+6.8$ percent while the richest loses -1.33; and across rent types the largest gains accrue to households holding *expensive* draws ($+8.05$ percent), the households the lock-in economy strands.

Table 7: Experiment A: removing rental-rate risk — tenure and welfare.

	Result
Aggregate ownership response (percentage points)	
market unit root removed (Shapley share)	+0.16
idiosyncratic dispersion removed (Shapley share)	+2.74
re-entry re-draw removed (Shapley share)	+0.38
all rental-rate risk removed	**+3.29**
pure-insurance margin (uncertainty only, dispersion kept)	−0.36
Welfare: consumption-equivalent value of removal (percent)	
mean	+1.08
p10 / p90 of the ex-post distribution	−9.4 / +4.7
poorest / richest lifetime-consumption quintile	+6.8/-1.33
cheap-draw / dear-draw rent type	+1.01 / +8.05

Notes. Preferences fixed at the estimates; each removal is mean-preserving and paired across worlds by common random numbers. Shapley shares attribute the total across the three components. The pure-insurance row freezes the idiosyncratic chain (no uncertainty) while keeping the cross-sectional dispersion of draws: the small negative value is the insurance motive; the large positive total is the lock-in value of the dispersion itself.

Lock-in and escape (A). The lock-in margin is directly visible in behavior: a renter in the priciest tercile of current draws buys at rate 0.51 against 0.29 in the cheapest — expensive draws push households into ownership, cheap draws hold them back. When draws are made portable, the mobility of pricey-draw renters falls from 0.39 to 0.15: a large share of renter mobility is escape — moving to re-draw the rent — and it disappears once the draw travels with the household.

Table 8: Experiment A′: lock-in in behavior — renter transitions by current rent draw.

Renter tercile of current draw	Buy rate	Move rate	Move rate, portable draws
Cheapest	0.29	0.16	0.20
Middle	0.38	0.31	0.19
Priciest	0.51	0.39	0.15

Notes. Three-year rates among renters, by tercile of the current idiosyncratic draw. Buying rises steeply in the draw — expensive draws push renters into ownership, cheap draws hold them back — and mobility among pricey draws collapses once draws are made portable: that mobility is escape, not preference.

Which risk does what (A″). Experiment A isolates rental-rate risk; Table 9 places it alongside the model's other two risk sources. We repeat the Shapley construction one level up: the three removable sources are now the full rental-rate bundle (market, idiosyncratic, and re-draw), house-price risk (prices frozen at their unconditional mean), and income risk (persistent and transitory innovations shut down, with income types and the age profile retained, so the column measures risk rather than heterogeneity). All 2^3 worlds are mean-preserving and simulated under com-

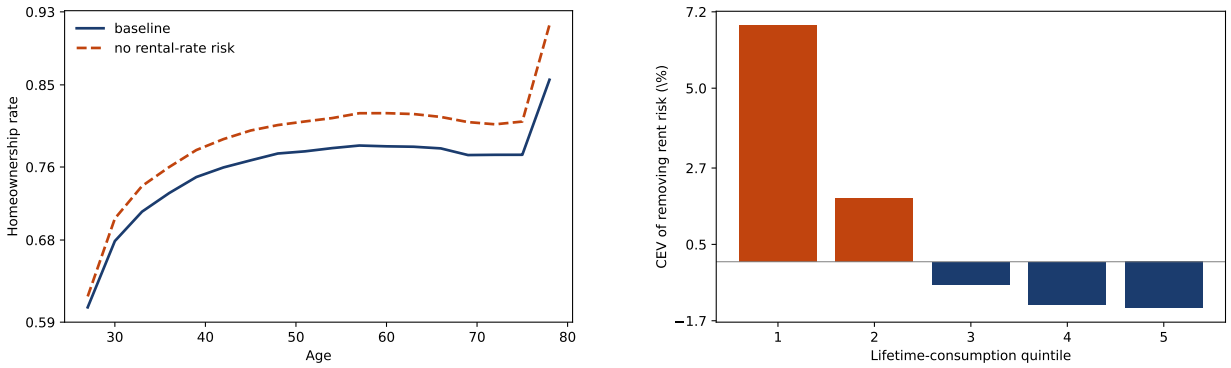
mon random numbers, and we decompose both the change in aggregate ownership and the mean consumption-equivalent gain from removing all risk. The result is a clean separation of labor: *each risk owns a different margin*. House-price risk owns the tenure margin — it accounts for +19.8 of the +23.7pp total, and removing it alone lifts ownership to 0.943, because a house whose resale value cannot fall is close to a dominant asset at these rent levels. Income risk owns the welfare margin — 58.7 of the 43.4 percentage points of consumption-equivalent gain, the familiar lifecycle result that uninsured persistent earnings risk is the first-order welfare burden, while its effect on tenure is small and *negative* (−0.93pp: precautionary wealth spills into house purchases, so removing income risk slightly lowers ownership). Rental-rate risk owns neither aggregate: +4.85pp of ownership and +0.63 points of welfare. Its action, as Experiment A shows, is distributional — it reallocates roughly eight percent of lifetime consumption across renters by draw while leaving the averages nearly unchanged. Two honest caveats. First, the house-price welfare share is negative (−16.0): with mean-reverting prices, "risk" bundles an exploitable buy-low option, and freezing prices removes those return opportunities along with the variance, so the tenure and welfare columns for housing move in opposite directions. Second, the income-risk welfare number (+61.6% when removed alone) prices the complete elimination of uninsured idiosyncratic earnings risk under $\gamma = 2.33$ — an intentionally extreme intervention, reported for scale rather than as a policy counterfactual.

Table 9: Experiment A″: which risk does what — Shapley decomposition across the three risk sources.

Risk source removed	Δ ownership (pp)	Share of welfare gain (CEV, %)
House-price risk	+19.8	−16.0
Rental-rate risk (all components)	+4.85	+0.63
Income risk (innovations only)	−0.93	+58.7
All three	**+23.7**	**+43.4**

Notes. Shapley attribution over the 2^3 mean-preserving, CRN-paired worlds; columns sum exactly to the all-risks-removed totals. Ownership responses are percentage points of the aggregate rate; welfare shares are percentage points of the mean consumption-equivalent gain from removing all risk. The negative house-price welfare share reflects that mean-reverting price *risk* bundles an exploitable buy-low option: freezing prices removes return opportunities along with variance.

Re-anchored world (B). Matching baseline ownership in a world without rental-rate risk requires *raising* the renter housing weight from 0.431 to 0.455: absent the lock-in claims that keep renters renting, rented housing must be made more attractive to hold the tenure split fixed. The re-anchoring confirms the decomposition's reading from the opposite direction.



(a) Ownership over the life cycle

(b) Welfare by consumption quintile

Figure 5: The decomposition. Left: homeownership by age, baseline (solid) versus a world with no rental-rate risk (dashed) — ownership is higher without rent risk at every age: the cheap-draw claims that rent risk creates hold renters back from buying. Right: consumption-equivalent value of removing all rental-rate risk by lifetime-consumption quintile — strongly bottom-loaded, with the poorest quintile gaining most and the top quintiles paying.

Table 10: Experiment B: the re-anchored world.

	Value
ψ_0 at the estimates	0.431
ψ_0 required to hold baseline ownership absent all rent risk	0.455

Notes. Removing all rental-rate risk and re-solving for the renter housing weight that restores baseline aggregate ownership. The required weight *rises*: without the lock-in claims that keep renters renting, rented housing must be made more attractive to hold the tenure split fixed — the mirror image of Experiment A.

Policy (C). Halving the down-payment requirement moves ownership by -0.02 points: at the adopted parametrization the constraint delays purchases briefly but does not bind in the aggregate. A mortgage subsidy that halves the borrowing wedge moves ownership by -0.04 points and carries a willingness-to-pay of $+1.48$ percent that we read as the value of an unfunded transfer, not a welfare verdict. Full lease portability — a stylized security-of-tenure policy — *raises* ownership by 0.82 points while cutting escape mobility (0.39 to 0.15 among pricey draws): making deals portable releases locked-in renters, some of whom buy. It delivers $+0.71$ percent of consumption on average and $+2.03$ percent to the poorest quintile — insurance the market does not price, provided without crowding out ownership.

Table 11: Experiment C: policy counterfactuals (partial equilibrium).

Policy	Δ own (pp)	Δ own, age 30 (pp)	CEV mean (%)	CEV Q1 (%)
Down payment 20 → 10%	-0.02	0.09	+1.26	+2.34
Mortgage wedge halved	-0.04	-0.07	+1.48	+1.68
Full lease portability	+0.82	+1.14	+0.71	+2.03

Notes. Each policy changes the household constraint set at fixed prices; the partial-equilibrium caveat applies throughout. The mortgage-subsidy willingness-to-pay is the value of an unfunded transfer, not a welfare verdict. Portability is the only instrument that both raises ownership and delivers bottom-loaded insurance.

5 Conclusion

Renting exposes a household to the future of a price; owning retires that exposure and, at the same time, forecloses the option of holding on to a cheap rental deal. This paper measures the price process at the level at which it constitutes risk — the rent an individual household pays for a given unit — and lets the measurement discipline what a life-cycle model says about tenure.

The measurement's central difficulty is that a household panel cannot, by itself, separate the persistence of rental-rate risk from permanent heterogeneity across households: the two are near-observationally-equivalent at the frequencies and spell lengths the PSID provides. Combining the panel with external rent indexes resolves the problem. Aggregate indexes, which carry no household fixed effect and are long, point-identify the persistence of the market component — at or near a unit root; netting that component out identifies the idiosyncratic remainder, whose permanent variance is interior rather than at a boundary; and the gap between new-tenant and sitting-tenant indexes measures the re-entry margin as a dispersed re-draw rather than a predictable discount. The estimated household process tracks the official index within region.

Carried through an estimated model, the process overturns the presumption the paper's question begins from: rental-rate risk does not push households into owning — it holds them back. Removing the risk raises aggregate homeownership by 3.29 percentage points, almost entirely through the idiosyncratic dispersion that turns cheap rental draws into claims worth keeping; the pure-insurance margin is small. The risk's consumption-equivalent burden is two-sided and bottom-loaded — largest for the poorest households and for those stranded on expensive draws — and the lock-in is visible in behavior, in buy rates that double across the draw distribution and in escape mobility that vanishes when deals are made portable. A security-of-tenure policy that grants that portability supplies the insurance directly and releases locked-in renters into ownership: tenant protection and homeownership emerge as complements, not rivals.

Three extensions follow. The restricted-geocode PSID, for which our application is pending, would replace the four census regions with metropolitan areas and sharpen the separation of market-level from unit-level rent risk. A mortgage amortization schedule would discipline the model's one remaining systematic misfit — owners who deleverage too fast and young house-

holds who buy somewhat too early. And endogenizing the rent and price processes would give the policy experiments general-equilibrium content that the present partial-equilibrium exercise deliberately leaves aside. The measurement, however, stands on its own: a model of tenure choice inherits its answer from the persistence of rental-rate risk, and that persistence is now identified rather than assumed.

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A Data Construction

The estimation sample is built from the PSID family files, 2001–2021, merged and deflated to constant dollars by the project's Stata pipeline. Heads of household aged 27–80 with positive family weight and tenure status owned or rented enter the panel: 53,432 head–wave observations from 9,841 households. Renters require monthly rent in a plausible range and positive real rent; owners require self-reported house value below the top-code and positive real value; both require room counts in $[1, 20]$. Rents are deflated by the CPI all-items-less-shelter index, region-matched, so that rent movements are measured against non-housing purchasing power rather than against a basket that contains shelter itself. First differences are taken over consecutive waves within household, restricted to spells in which the household reports not having moved and the reported room count is unchanged — the *strict stayer* sample of 3,823 rent first-differences from 21,319 renter observations and 6,108 households.

Three external series enter. The CPI rent of primary residence by census region (monthly, from December 1977) supplies the market-persistence estimate and the region-by-year netting; the region-matched all-items-less-shelter series is the deflator. The BLS new-tenant and all-tenant repeat-rent research indexes (quarterly; all-tenant from 1999, new-tenant from 2005) supply the re-entry margin. The Zillow observed rent index (metropolitan, from 2015) is reserved for cross-metropolitan dispersion once restricted-geocode PSID data permit a metropolitan match. External files are downloaded once and frozen with checksums.

Table 12 reports summary statistics by tenure. The contrasts that discipline the model are visible in the raw data: the median renter holds \$3,500 of net worth against the median owner's \$201,000, renters occupy four rooms against owners' seven — the quality gap between the stocks that motivates the renter housing weight ψ_0 — and half of renters move between consecutive waves against one in seven owners.

Table 12: PSID estimation sample: summary statistics by tenure.

	Renters	Owners
Head–wave observations	28998	53005
Households	7917	9877
Age (median)	41	53
Total family income (median, 2010$)	40000	78000
Net worth incl. home equity (median, 2010$)	3500	201000
Net worth / income (median)	0.11	2.71
Annual rent (median, 2010$)	8400	—
Rent / income (median)	0.21	—
House value (median, 2010$)	—	200000
House value / income (median)	—	2.48
Home equity / income (median)	—	1.29
Share with a mortgage	—	0.69
Rooms (median)	4	7
Moved since last wave (share)	0.49	0.14

Notes. Heads of household aged 27–80, PSID family files 2001–2021, family weights. Dollar values in 2010 dollars. Net worth is the PSID imputed wealth-with-equity concept (WEALTH2); the mortgage share is the share of owners whose self-reported house value exceeds imputed home equity by more than $1,000. Sample construction in Appendix A.

B External Market Persistence: Method

For each census region and the nation we form an annual real log rent index as the January–June average of the monthly index (numerator and region-matched deflator), matching the PSID's spring interview timing, and estimate an annual AR(1) with a linear trend. Because least squares is biased toward stationarity near a unit root, we report the exactly median-unbiased estimate and a 90 percent quantile-unbiased interval obtained by inverting the simulated median and quantile functions of the least-squares estimator at the exact sample size (Andrews, 1993), using 20,000 replications per grid point. We test for a unit root by the augmented Dickey–Fuller statistic under a decision rule fixed before estimation: if the unit root is rejected at five percent we report the median-unbiased persistence with its interval; if it is not rejected, the market component enters the model as a unit root with drift and persistence is reported as an interval. The governing series is the national index over the regional sample window, chosen in advance; its unit root is not rejected ($p = 0.644$), and the three-year innovation standard deviation is 0.038. Only the Northeast rejects the unit root, and enters the regional evidence as a trend-stationary contrast.

C Identification of the Error-Components Model

Within stayer spells the residual rent obeys $u_{it} = f_i + z_{it} + m_{it}$, where f_i is a permanent household-and-unit effect with variance σ_f^2, $z_{it} = \rho z_{i,t-1} + \eta_{it}$ is a persistent shock with stationary variance

σ_z^2 and innovation variance $\sigma_\eta^2 = \sigma_z^2(1-\rho^2)$, and m_{it} is i.i.d. measurement error with variance σ_m^2. Differencing removes f_i:

$$\begin{aligned} \mathrm{Var}(\Delta u) &= 2\sigma_z^2(1-\rho) + 2\sigma_m^2, & \mathrm{Cov}(\Delta u_t, \Delta u_{t+1}) &= -\sigma_z^2(1-\rho)^2 - \sigma_m^2, \\ \mathrm{Cov}(\Delta u_t, \Delta u_{t+2}) &= -\sigma_z^2\,\rho\,(1-\rho)^2, & \mathrm{Cov}(u_t, u_{t+k}) &= \sigma_f^2 + \sigma_z^2\rho^k \quad (k \geq 1), \end{aligned}$$

with σ_m^2 loading only on the level variance. In levels, the autocovariance is $\sigma_f^2 + \sigma_z^2\rho^k$: a tail that decays slowly is consistent with a large σ_f^2 and small persistence, with a small σ_f^2 and ρ near one, or with any mixture — the near-non-identification the external step resolves. We net the estimated market component out of household rents through region-by-year fixed effects in the hedonic, then estimate $\theta = (\rho, \sigma_\eta^2, \sigma_f^2, \sigma_m^2)$ by equal-weighted minimum distance on seven autocovariance moments (first-difference variance and lags one and two; level variance and lags one, two, three), with $\sigma_f^2 \geq 0$ imposed inside the solve. Equal weighting is used in place of the efficient weighting matrix, which finite-sample evidence shows is severely biased for covariance structures (Altonji and Segal, 1996). Standard errors are a household block bootstrap (999 replications), and a Hall–Horowitz recentered bootstrap test of the two overidentifying moments does not reject the specification ($p = 0.122$). Because any deflator that varies only at the region-year level is absorbed by the fixed effects, the moments are numerically invariant to the deflator, a check that holds to machine precision.

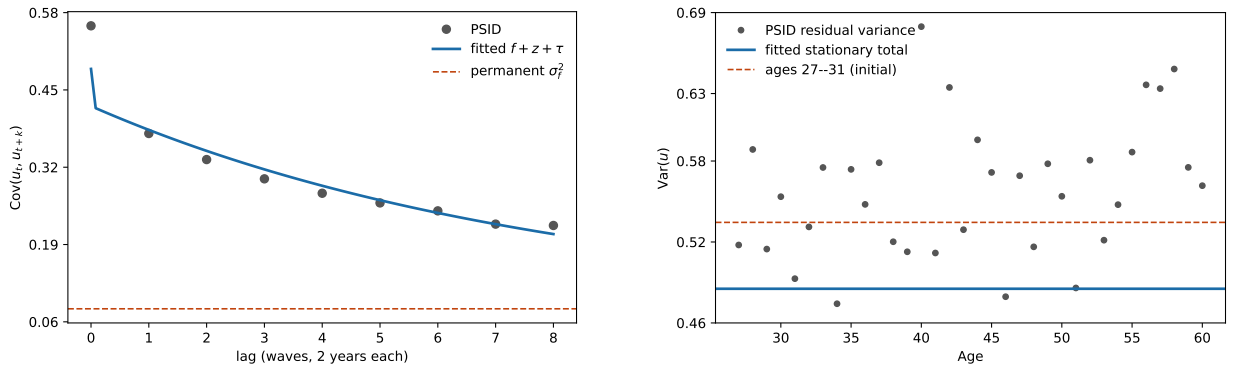
D Income Process Estimation

The income process is estimated on the PSID panel by the same error-components method used for rents, on a sample built to the conventions of the income-dynamics literature: household heads employed at interview, aged 27–60, with *family* labor income — head plus spouse, in 2010 dollars, at least \$1,000 — below the top-code; first-difference pairs with residual changes above four log points are dropped (31950 head–wave observations, 7647 households). Family rather than head-only earnings is the concept the model's budget constraint carries, and it matters: spousal earnings smooth household resources, so head-only income would overstate the risk the household faces. Log real family labor income is residualized on a hedonic of age, age squared, education, sex, race, region, family size, and year; the deterministic age-earnings profile χ_t in (3) is the fitted age effect, rising to 1.62 times the age-27 level at 51; Figure 7 plots the raw PSID profile against both the hedonic age effect and the binned profile the model consumes.

The residual is decomposed as $u_{it} = f_i + z_{it} + \tau_{it}$: a permanent household component, a persistent autoregression, and a transitory-plus-measurement-error composite, fit by equal-weighted minimum distance to the first-difference variance and autocovariances and to level autocovariances out to lag eight — sixteen years, a tail long enough to separate permanent heterogeneity from persistence, which is where the short-tail rent decomposition needs external data and the income decomposition does not. The estimates: persistence 0.829 per three-year period (90 per-

cent bootstrap interval $[0.718, 0.904]$), innovation standard deviation 0.326 $[0.270, 0.368]$, permanent variance 0.082 $[0.000, 0.178]$, and transitory composite standard deviation 0.250. A profile of the objective over the persistence grid is single-troughed — the long tail point-identifies the split. Figure 6 shows the fit: the level-autocovariance tail against the fitted decomposition, whose asymptote is the permanent variance, and the age profile of the residual variance, whose age-27–31 value (0.535) initializes the model's persistent state. Table 13 reports the income and house-price estimates together.

The model consumes the decomposition as identity and risk separately: the permanent component enters as three mean-one income types — known to the household, carrying no precautionary motive — while only the persistent innovation and the transitory composite enter as shocks. This distinction, identity versus uncertainty, is the income-side analogue of the paper's rent-side distinction between heterogeneity and risk, and it disciplines precautionary wealth: treating the permanent spread as a shock, as a specification without types implicitly does, overstates the buffer stocks households hold against it.



(a) Level autocovariances and fit

(b) Residual variance by age

Figure 6: Income process: data and estimated profiles. Left: level autocovariances of residual log family labor income by lag (markers) against the fitted $f + z + \tau$ decomposition (line); the dashed line is the permanent variance σ_f^2, the tail's asymptote. Right: residual variance by age (markers), the fitted stationary total (solid), and the age-27–31 value used to initialize the model's persistent state (dashed).

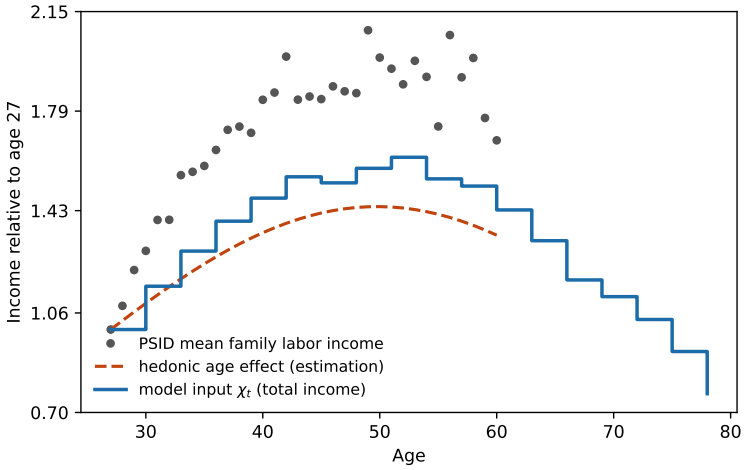


Figure 7: The age-earnings hump: PSID versus the estimated and fed profiles. Markers: weighted mean family labor income by age in the estimation sample (employed heads, 27–60), relative to age 27. Dashed: the hedonic quadratic age effect from the income estimation, same normalization. Solid step: the profile χ_t fed to the model — weighted mean total family income by three-year bin over the full 27–78 sample, peaking at 1.62 times the age-27 level at 51 and declining thereafter as retirement income replaces earnings. The raw profile is steeper than either fitted object because education, family-size, and year effects that covary with age load into the raw means; the hedonic age effect nets them out, and χ_t averages over total income including transfers.

E House-Price Process Estimation

House prices are estimated from PSID owners' self-reported house values, over stayer spells in which the reported value refers to the same unit, again by the error-components method. The log real house value decomposes into a permanent unit-and-owner fixed effect, a persistent autoregressive component, and measurement error. Unlike rents, house values *do* separate the fixed effect from persistence within the panel: the level autocovariance tail decays fast enough that the fixed-effect variance is identified interior to its boundary, $\sigma_f^2 = 0.215$ $[0.000, 0.276]$, with a three-year autocorrelation of 0.650 $[0.401, 0.879]$ and an innovation standard deviation of 0.327 per period $[0.287, 0.376]$. This identification asymmetry — house values identified within the panel, rents not — is exactly what motivates the external-index strategy for rents in Section 2: the rent tail is flat where the house-value tail decays. Figure 8 plots the estimated versus empirical level and first-difference variance profiles by age, the moments the estimator targets.

Table 13: Income and house-price processes: error-components estimates.

	Persistence ρ	Innovation σ_η	Fixed effect σ_f^2
Log income	0.829 [0.718, 0.904]	0.326 [0.270, 0.368]	0.082
Log house price	0.650 [0.401, 0.879]	0.327 [0.287, 0.376]	0.215 [0.000, 0.276]

Notes. Estimated on PSID stayer spells by the error-components method of Appendix D–E; three-year model frequency. Brackets are 90 percent household-block-bootstrap intervals. Income adds a transitory component and the deterministic age profile χ_t; house values, unlike rents, separate the fixed effect from persistence within the panel.

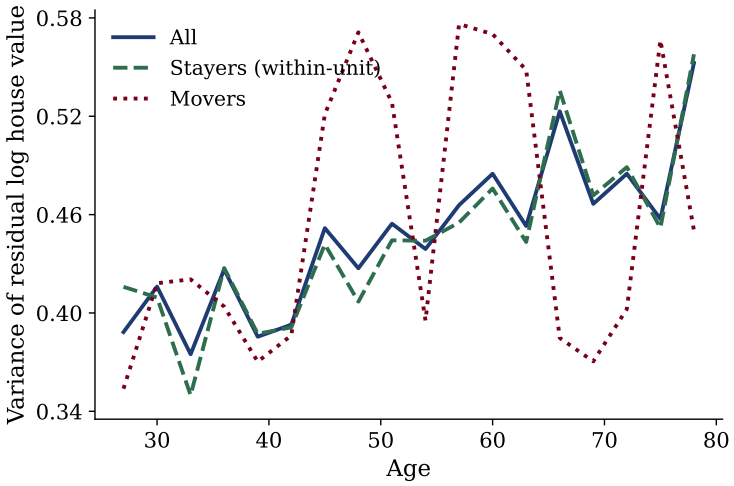


Figure 8: House-value variance profiles: model-implied (lines) versus PSID (markers), level and first-difference variances by age. The fitted profiles are the moments the error-components estimator targets.

F The Move-Contingent Re-Draw

The idiosyncratic rent state z evolves within a tenancy by its estimated autoregression. When a household enters a new rental unit, z is partially re-drawn: it follows a transition with correlation ρ_{move} to its previous value, on the same grid and with the same stationary distribution, so the re-draw preserves the cross-sectional distribution of rents. We set ρ_{move} from the certified data. Let the measurement-error-corrected variance of rent changes across moves be V_{move} and the stationary idiosyncratic variance be σ_z^2. Under the partial re-draw, $V_{\text{move}} = 2\sigma_z^2(1 - \rho_{\text{move}})$, so $\rho_{\text{move}} = 1 - V_{\text{move}}/(2\sigma_z^2) = 0.446$. A fully independent re-draw ($\rho_{\text{move}} = 0$) would imply mover variance $2\sigma_z^2$, which exceeds the observed value; the data therefore call for a *partial* re-draw. The average level step at a move — the sitting-tenant discount forfeited — is set to zero by the symmetry of the re-draw, consistent with the small and sign-ambiguous mover-mean estimate.

G Computation

The model is solved by backward induction with exact discrete maximization over the (a', h') grid at each state, with no smoothing of the housing-size choice; only the binary tenure choice carries the Type-I extreme-value taste shock. The asset–housing grid is $(21, 7)$, with zero included as an explicit asset point so that the renter no-borrowing constraint binds at exactly zero rather than at the nearest positive grid point — an error that would otherwise force saving on every renter and move the feasibility boundary across grid refinements. Income, rent, and price states are discretized by Rouwenhorst, and the market rent component is an age-expanding three-point grid whose level multipliers are renormalized to mean one at every age, so that shutting the component down in counterfactuals is mean-preserving by construction. States at which no choice affords positive consumption carry a large finite penalty that is capped before entering continuation-value expectations, so infeasibility never contaminates the value function's magnitudes; all reported results are computed at the reference grid, the same resolution at which parameters were selected. Table 14 reports the full parameter scan behind the estimates: every Sobol point evaluated, its loss, and its implied aggregate ownership and life-cycle gradient. Two features deserve note. The gradient is positive at every point in the box, so the rising profile is a structural property of the model rather than a product of parameter choice; and the loss surface is visibly flat across several parameter mixes near the optimum, an identification limit of the five-moment set that the full table makes inspectable rather than hidden.

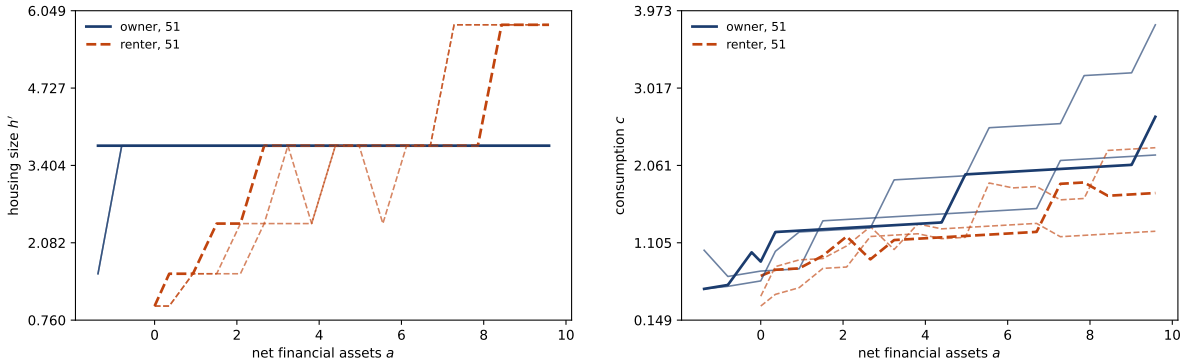
Table 14: The global parameter scan: all evaluated points.

Rank	ψ_0	θ_B	γ	ν	Loss	Ownership	Age gradient
adopted	0.431	2.15	2.33	0.60	0.187	0.758	+0.097
2	0.899	1.75	1.90	0.14	0.216	0.583	+0.055
3	0.497	0.36	2.56	0.52	0.217	0.742	+0.086
4	0.655	0.93	2.10	0.38	0.220	0.677	+0.074
5	0.550	2.32	1.45	0.28	0.430	0.889	+0.091
6	0.614	1.57	1.69	0.22	0.450	0.901	+0.100
7	0.834	0.75	1.20	0.08	0.525	0.894	+0.089
8	0.720	2.99	2.82	0.46	0.529	0.577	+0.047
9	0.627	1.88	2.98	0.10	0.651	0.965	+0.108
10	0.771	0.49	1.73	0.55	1.404	0.436	+0.034

Notes. Scrambled Sobol draws over $\psi_0 \in [0.4, 0.9]$, $\theta_B \in [0.2, 3]$, $\gamma \in [1.1, 3]$, $\nu \in [0.05, 0.6]$, every point evaluated at the reference grid with 50,000 simulated lives; the adopted point re-evaluated at 100,000. The loss is the equal-weighted sum of squared relative gaps on the five targeted moments. The age gradient (ownership at 69 minus at 30) is positive at every point: the life-cycle rise is a property of the model, not of the chosen parameters.

Every solution passes hard-asserted property tests before any result is used: value functions monotone in assets and in transitory income; the entering-state owner-minus-renter value gap

positive and globally declining in assets, as concavity in wealth requires; and the renter's tenure-choice probability non-decreasing in assets inside the feasible region, with every local exception co-located with a switch in the discrete housing-size policy. Welfare is computed with Bellman timing — flow utility over chosen quantities and the bequest at the chosen terminal position — and consumption-equivalent variations are solved per simulated agent by bisection, with agents paired across worlds by common random numbers. Figures 9 and 10 display the housing, consumption, and saving policies at the estimated parameters.



(a) Housing size h'

(b) Consumption c

Figure 9: Housing and consumption policies by tenure (solid owner, dashed renter), ages 30/51/72, median shocks, middle rent type.

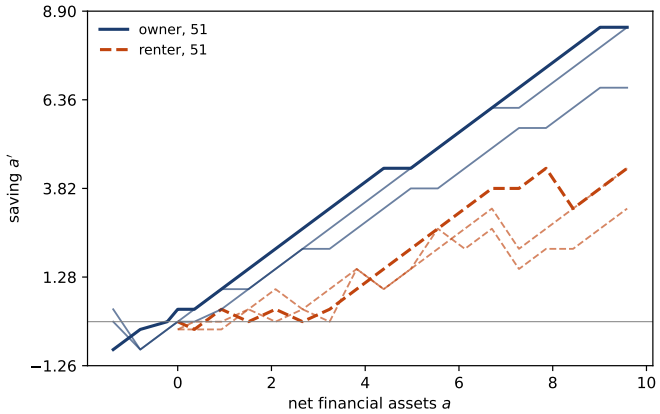


Figure 10: Saving policy a' by tenure; $a' < 0$ is mortgage debt, borrowed at the mortgage rate above the savings rate. The borrowing region is used by owners only — renters face $a' \geq 0$.